Who Bears Flood Risk? Evidence from Mortgage Markets in Florida∗

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Government-provided flood insurance contracts have strict coverage limits, leaving some households under-insured against flood risk. This paper exploits these strict coverage limits as well as staggered flood map updates to show that mortgage lenders screen for uninsurable flood risk by requiring lower loan-to-value ratios at origination. This credit rationing leads delinquency rates to equalize inside and outside of flood zones, and shifts the composition of mortgage borrowers in flood zones towards richer and higher credit quality individuals. I conclude that lenders reduce credit supply when they retain residual uninsured exposures to flood risk, which has distributional consequences for flood zones. (*JEL* JEL G00, G21, G22, E6, Q54, R00)

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In the United States, 35.8 million homes with a combined market value of $6.6 trillion are exposed to natural disaster risk (RealtyTrac, 2015). Floods are the most costly natural disasters, having caused hundreds of billions of dollars in damage over the past few decades (Figure 1). These costs are expected to rise further, with climate change bringing rising sea levels, heavier rains, and stronger hurricanes (Davenport, Burke, and Diffenbaugh, 2021; IPCC, 2023). By some estimates, flood-related property damage could increase by more than 60% over the next 30 years due to climate change (Duguid, 2021).

Given the large scale of potential exposures, financial regulators worry that banks may neglect or misprice flood risk, possibly threatening

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financial stability.[1] Mortgage lenders originate more than $200 billion annually in flood zones, representing roughly 10% of total bank equity (Ouazad, 2020). Bank regulators in Canada, France and the United Kingdom are already conducting climate-related stress tests, and the U.S. Federal Reserve is considering a similar approach (Brainard, 2021). Similarly, climate activists worry about over-development in high risk coastal areas and the role played by financial markets. If lenders neglect to incorporate flood risk into mortgage terms, then borrowers may not bear the true costs of flood risk and have incentivizes to move into high risk areas. Despite intense policy interest, there are few empirical studies on the distribution of flood risk in the U.S. economy.

In the market for flood risk, there are three primary agents: households, mortgage lenders, and the government. Households bear flood risk because floods can cause property damage and reduce home values. Lenders bear flood risk because households purchase their homes using mortgages. The mortgage's loan-to-value (LTV) ratio at origination is a key determinant of how this risk is shared between them. Households and banks may insure flood risk through the government-run National Flood Insurance Program (NFIP).[2] Flood insurance payments can be used to repair damaged homes, changing household default incentives and lender collateral risk exposures. Flood insurance is required by law for most mortgage borrowers in flood zones. Importantly, these government flood insurance are incomplete, leaving residual uninsurable flood risk borne by households and lenders.

This paper uses a novel dataset and two quasi-exogenous shocks to study how flood risk exposures are distributed across banks, households, and the government flood insurer through residential mortgage contracts in Florida. Data is a key challenge for estimating the impact of flood risk on mortgage markets for two reasons. While current flood maps are available digitally as a database from FEMA, historical flood maps are not, making it difficult to identify a mortgage's flood zone classification at origination. A second challenge is that standard mortgage datasets do not include property characteristics, such as the address. I address these challenges by digitizing historical government flood maps that were previously unavailable, and geo-locating individual mortgages using a fuzzy merge of mortgage and property deeds data. The final dataset identifies each mortgage's flood zone classification at origination.

[1] Fed Governor Lael Brainard says in her December 18, 2020 speech: "It is vitally important to strengthen the U.S. financial system to meet the challenge of climate change... We are already seeing elevated financial losses associated with an increased frequency and intensity of extreme weather events... Mortgages in coastal areas are vulnerable to hurricanes and sea level rise." (Brainard, 2020)

[2] Flood-related damages are carved out of traditional homeowners insurance contracts. The NFIP is the dominant player in the market for flood insurance. Private flood insurance, both primary and supplementary top-up coverage, is rare in my sample period.

I supplement the flood risk data with information from public and commercial sources, allowing me to obtain a a rich set of borrower and property-level characteristics including applicant income, credit score, property value, replacement cost, and securitization status, as well as the subsequent performance of the mortgage. The data I assemble covers single-family purchase mortgages in Florida originated between 2010 and 2016.

Even with this loan-level data, estimating the causal effect of flood risk poses a significant identification challenge because flood zones have unobserved amenities (e.g. water views) which may attract wealthier borrowers that select different mortgage contracts. I consider two natural experiments to address this omitted variable. The first natural experiment exploits strict flood insurance coverage limits that exogenously change flood risk exposures around the cap, and the second natural experiment leverages the release of new flood maps that change lender risk assessments while keeping underlying amenities fixed. Together, both natural experiments offer complementary but distinct insights into climate adaptation in mortgage markets, highlighting the importance of both comprehensive insurance availability and changing lender risk assessments in the face of new information about a dynamic and evolving climate change process.

Using my novel dataset and these two sources of plausibly exogenous variation, I obtain three key findings. First, lenders require lower LTV ratios (higher down payments) from borrowers in flood zones that are under-insured against flood risk. Second, lower LTV ratios at origination protect lenders against the delinquency risk associated with uninsured flood risk. I find that the LTV adjustment leads delinquency rates to be roughly equal for both fully insured and under-insured homes in and out of flood zones. Third, reduced credit supply has distributional effects, shifting the composition in risky areas to richer, higher credit quality borrowers. The results suggest that higher down payment requirements deter more liquidity constrained individuals from obtaining mortgages in high risk areas.

I begin the paper with novel descriptive facts on the characteristics of households and mortgage contracts by flood zone. I find that households in flood zones tend to be of higher credit quality, with higher incomes, property values, and credit scores, and lower LTV ratios. The LTV result holds even after controlling for borrower credit score, income, property value, and ZIP-year fixed effects. With the same set of controls and fixed effects, both interest rates and delinquency rates are equal in and out of flood zones. These results suggest that flood zones are advantageously selected, with mortgage contracts that are less risky on average from the perspective of the lender.

The fact that LTV ratios are significantly lower in flood zones raises the possibility that lenders may restrict credit supply in flood zone to manage flood risk exposures. However, the descriptive results preclude a causal interpretation because of an important omitted variable: unobserved amenities. The econometric issue is that flood zones are disproportionately likely to have water views which may attract wealthier borrowers who tend to have higher down-payment mortgages. I use two empirical approaches to address this omitted variable and tease apart credit supply from credit demand.

In my first empirical approach, I exploit strict flood insurance coverage limits that leave some borrowers in flood zones under-insured. Flood insurance is required by law for most mortgage borrowers in flood zones.[3] Government flood insurance has a 95% market share, and these contracts only cover up to $250,000 in flood damage. Importantly, there is limited availability of private top-up insurance for borrowers in high risk flood zones in Florida. As a result, smaller homes are fully insurable, and larger homes in flood zones are under-insured. This feature of flood insurance allows comparisons of mortgage contracts by flood zone status and by how much of the home's replacement cost can be insured against flood risk. The identifying assumption in this approach is that the unobserved amenities of flood zones are unlikely to discontinuously change at the flood insurance coverage limit. I find that LTV ratios are only lower in flood zones when homes are under-insured. That is, flood zone status has no effect on LTV ratios when the loan can be completely insured. However, the relationship between flood zone status and LTV ratios changes at the insurance coverage limit. Above that threshold, LTV ratios are significantly lower in flood zones, and LTV ratios are reduced more in flood zones when a greater share of the home is uninsurable. These patterns are also robust to controlling for borrower credit score and income, consistent with a credit supply interpretation. There is no statistically or economically significant adjustment of interest rates, and delinquency rates are roughly equal in and out of flood zones for each replacement cost category.

In my second empirical approach, I exploit periodic flood map updates which change risk assessments while keeping underlying amenities fixed. Government flood maps produced by FEMA are the most widely used measures of flood risk (Kousky et al., 2018).[4] All zip codes in a county

[3] Compliance with the requirement is less likely to be an issue. FEMA's calculations suggest that, in 2015, 60% of mortgage borrowers in high risk flood zones had a flood insurance policy, with these rates likely being higher in Florida, which has the highest take-up rates nationally (FEMA, 2018).

[4] Today there are many commercial providers of flood risk information, such as the First Street Foundation and CoreLogic, but during my sample period of 2010-2016, there were few alternatives to the government flood maps. I focus on government flood maps since that

receive updated flood maps at the same time, although some zip codes have expanded boundaries while others have unchanged or contracted boundaries. I find that zip codes where flood zones expand have a reduction in LTV ratios by 2 percentage points. Average interest rates do not significantly change, and delinquency rates decline significantly by 1 percentage point. The flood insurance coverage limit also plays an important role. I find that LTVs reductions following the remappings are concentrated among higher credit risk borrowers with uninsurable flood risk. This supports a credit supply interpretation of the results, since such borrowers are unlikely to prefer higher down payment contracts. I also find that the composition of mortgage borrowers changes after these flood zone expansions, with both mortgagor incomes and credit scores significantly increasing.

Taken together, these results show that mortgages in flood zones have lower LTV ratios, which reduces delinquencies and changes the composition towards higher income and higher credit score borrowers. This demonstrates the importance of insurance and mortgage markets for determining the allocation of households to high risk areas.

This paper makes three contributions to the literature. First, this paper contributes to the growing literature related to financial impacts of natural disasters on mortgage markets. This literature has primarily explored the causal effect of climate risk by examining ex-post effects of realized natural disasters on mortgage performance (Bleemer and van der Klaauw, 2019; Billings, Gallagher, and Ricketts, 2019; Issler et al., 2019; Kousky, Palim, and Pan, 2020), mortgage origination (Cortés and Strahan, 2017; Gropp, Noth, and Schüwer, 2019;), and mortgage securitization (Ouazad and Kahn, 2021). So far the literature has found transient and limited effects of realized disasters on mortgage pricing (Garbarino and Guin, 2021; Correa et al., 2020). A key challenge to this ex-post approach is that it is difficult to ascertain where and how much risk assessments should change in response to the experience of a natural disaster. My contribution to this literature is to consider new sources of plausibly exogenous variation in flood risk that stem from *ex-ante* measures. Variation from flood insurance contracts and updated flood maps allow for directly measuring the independent effect of fundamental risk exposures and assessments on mortgage terms for a given location. Using these sources of variation, I show that mortgage terms adjust to uninsurable flood risk through lower loan-to-value ratios

In a contemporaneous paper, Santos and Blickle (2022) also use flood map updates to study the effect of required flood insurance on

information could feasibly be in lenders' information sets when mortgages are originated. In addition, private sector measures of flood risk, such as those from CoreLogic or the First Street Foundation, use the FEMA data and flood maps as an input.

borrower credit demand. Relative to their study, this paper focuses on identifying credit supply effects by examining updated flood maps along with variation in the insurability of a property.

Second, this paper draws from the large literature on household default and negative equity. This literature explains why requiring lower LTV ratios at origination can reduce delinquency risk for lenders. The literature describes household default behavior using either a pure "strategic default" model where default is entirely triggered by negative equity (Foster and Van Order, 1984) or a "double-trigger" model where household default is triggered by both negative equity and cash-flow shocks (Foote and Willen, 2018). Much of the literature finds limited evidence of pure strategic default behavior.[5] Importantly, in both the pure strategic default and double-trigger models, negative equity is a necessary condition of default, since borrowers with positive equity are better off repaying loans using the proceeds from selling the home and keeping any remaining balance.[6] LTV ratios at origination affect the borrower's equity position throughout the life of the loan and therefore serve as a useful tool for managing the collateral exposure to uninsurable flood risk.

Lastly, this paper joins a large literature which shows that the ability to transfer risk changes lender screening incentives. Downing, Jaffee, and Wallace (2009), Keys et al. (2010), Purnanandam (2011), and Keys, Seru, and Vig (2012) show that lenders securitize mortgages with the government-sponsored enterprises that are of lower credit quality than the ones they retain on balance sheet. Similarly, Campbell and Dietrich (1983) and Park (2016) find that government and private mortgage insurance diminish lender underwriting standards at origination. I contribute to this literature by showing a different source of moral hazard stemming from the government flood insurance program, which also reduces lender screening incentives for fully insurable risks.

1. Hypothesis Development

In this section, I consider a conceptual framework to derive three hypotheses of how banks should manage their flood risk exposure which can be tested in the data. The stylized model in Appendix B formalizes these intuitions.

The conceptual framework considers a borrower who has decided to purchase a home and then applies to a bank for mortgage financing.

[5] See, for example, Scharlemann and Shore (2016); Bhutta, Dokko, and Shan (2017); Fuster and Willen (2017); Gerardi et al. (2018); Ganong and Noel (2020).

[6] Low (2018) and Ganong and Noel (2020) note that frictions in housing markets can make it difficult to sell homes, explaining observed defaults from borrowers with positive equity experiencing cash-flow shocks.

In the two-period model, the mortgage is originated in the first period and repaid in the second period. Between those two periods, a flood can occur with some positive probability. A flood causes some property damage, where property damage follows some probability distribution and impacts the second-period liquidation value of the home. After the flood, the household can choose to repay the loan or default. The borrower incurs some loss of utility from defaulting, reflecting the costs associated with financial distress. The probability of a flood, the distribution of flood damage, and default costs are known to both the borrower and the bank. Both the borrower and the bank are risk-neutral, but there are gains from trade because the lender is assumed to be more patient than the borrower.

In the event of a flood, the borrower subsequently decides whether to default or repay the loan. The borrower's default rule will compare the outstanding balance of the mortgage with the value of the flooded property and the costs of default. If the property is worth less than what is owed to the banks and the costs of default, the borrower optimally defaults. The default rule is consistent with both the strategic default (Foster and Van Order, 1984) and double trigger (Foote and Willen, 2018) models, since in both models negative equity is a necessary condition of default. Given the default rule, the household will maximize expected utility subject to a lender zero profit constraint.

In the model, lenders change mortgage terms in flood zones to reduce the borrower's incentive to strategically default by requiring more borrower equity (skin-in-the-game). If default did not depend on loan sizes, then lenders could simply increase interest rates to manage flood risk. However, in this setting, a higher interest rate without a smaller loan size increases the amount owed to the bank, thereby increasing the incentive to default after a flood. While rationing credit lowers strategic default motives and offers more protection for the lender, doing so is not free – it comes at the cost of reduced payments from borrowers in flood zones who do not experience a flood but were rationed.

One approach lenders could take would be to obtain just enough "protection" from lower LTVs to exactly offset the flood risk they are exposed to by lending in flood zones. Such an adjustment rule can be rationalized by a simpler version of the model where flood risk is binary, and lenders adjust mortgage terms until they are indifferent between lending in and out of flood zones. Under such an adjustment rule, we would observe in the data that loan-to-value ratios are lower in flood zones, and that lending in flood zones should become no riskier than lending outside of flood zone. That is, lenders lower LTVs just enough that the delinquency risk in flood zones is, on average, the same as the delinquency risk outside of flood zones.

This conceptual framework leads to the following three hypotheses:

Hypothesis 1: LTV ratios are lower in flood zones.
Hypothesis 2: Interest rates are higher in flood zones.
Hypothesis 3: Delinquencies are the same in and out of flood zones.

These three hypotheses will frame the empirical tests throughout the rest of the paper.

2. Institutional Setting, Data, and Descriptive Evidence

U.S. mortgage markets are heavily regulated, with special provisions that influence how lenders can manage flood risk in particular. FEMA, a government entity, is the dominant provider of flood insurance. FEMA's flood zone designations determine flood insurance availability and pricing, with regulatory consequences for mortgage borrowers and lenders. To motivate my empirical approaches, I describe the key institutional features and related datasets here. Additional information on the institutional setting and data the can be found in Appendix Sections C and D.

2.1 Institutional Details

The National Flood Insurance Program (NFIP) was established in 1968 to provide insurance that had been historically unavailable through the private sector. Flood risk is carved out of standard homeowners insurance, and until very recently private flood insurance was rare (Kousky et al., 2018). FEMA administers the flood insurance program and today, the NFIP covers 95% of all residential flood insurance policies in the United States.

Flood Maps: To administer the insurance program, FEMA produces flood maps that define its classification of flood risk, with homes located in high risk flood zones facing higher flood insurance premiums.[7] FEMA's flood maps are the most widely used measures of flood risk (Kousky et al., 2018). FEMA is required to periodically conduct new engineering studies to determine whether flood maps should be revised; this process can lead to the release of new maps, which change the boundaries of flood zones.

Mandatory Purchase Requirement: FEMA's flood zone designations determine flood insurance premiums and have regulatory consequences for mortgage borrowers and lenders. By law, most mortgage lenders must require borrowers in high risk flood zones to purchase flood insurance. Specifically, the requirement stipulates that federal agencies, federally regulated lending institutions, and the government-sponsored enterprises (GSEs) must require borrowers in

[7] In Florida, moving from a low or moderate risk to a high risk classification leads to a $400 increase in annual premiums on average.

high risk flood zones to purchase flood insurance for every mortgage that these entities originate, guarantee, or purchase. When mortgage borrowers purchase flood insurance, their policy includes the name of their lender, and insurance claim checks are written to both the borrower and the lender. In addition, often times flood insurance payments are handled by mortgage servicers through escrow accounts.

Flood Insurance Coverage Limit: A key feature of the federal flood insurance program is that NFIP coverage is capped at $250,000 for residential structures. The private markets do provide excess flood coverage beyond the NFIP limit, but the availability of such products is limited for properties in high risk flood zones, especially in the states of Florida and Louisiana (Wells, 2006; Goldberg, 2005). This means that Florida households with replacement costs that exceed $250,000 will be under-insured against flood risk.

Further details on the institutional environment are avalable in Appendix Section C.

2.2 Data

The paper's empirical analysis considers how mortgage terms vary with flood zone classification and insurability. To do so, I construct a novel data set that combines geo-spatial data from flood maps with geo-located data on mortgage characteristics and performance. The final dataset focuses on the state of Florida and spans 2010-2016. The sample is restricted to purchase mortgages for single-family homes. Further details on the datasets and merge procedure are available in Appendix Section D.

FEMA Flood Maps: My primary measure of flood risk comes from FEMA's flood maps. In Florida, the latest flood maps and some historical ones can be downloaded directly from FEMA's Map Service Center as a geo-spatial database called the National Flood Hazard Layer. The digitized maps can be downloaded for an entire county, including all the individual maps for each community in that county. I obtain archived FEMA flood maps from geospatial libraries at Princeton, University of Texas, University of Florida, Harvard, and Berkeley, which saved snapshots of FEMA's floodmaps for various counties in Florida from 1996, 2001-2009, and 2011.[8] Lastly, I use tables from FEMA's Flood Insurance Studies, which accompany map updates, to compile the introduction and revision dates of each community's flood map.

[8] When I asked FEMA, they said archived flood maps were unavailable to be shared, which is why I compiled snapshots from these alternate sources. For a handful of counties, I manually digitized the floodmaps for these counties using PDFs of the historic map boundaries for each community to obtain maps that would be valid at the beginning of my sample.

Black Knight McDash: Black Knight is a private company that produces the McDash dataset, a comprehensive, loan-level dataset on mortgages that includes information on mortgage characteristics, borrower characteristics, and mortgage performance. The data is compiled from mortgage servicers and accounts for approximately two-thirds of the overall mortgage market. Mortgage characteristics in the data include the origination month, interest rate, LTV ratio, debt-to-income ratio, maturity, property value, and type of mortgage (e.g. FHA, VA, Jumbo, etc.). The data also include select borrower characteristics such as credit score. Location information is limited to 3-digit or 5-digit zip codes. Importantly, this data includes information on the performance of the mortgage from origination to its final payment. This includes whether the mortgage is current or in delinquency status, as well as events such as prepayment, default or foreclosure.

Home Mortgage Disclosure Act (HMDA): I obtain additional mortgage characteristics from the Home Mortgage Disclosure Act, which is administrative data on the near-universe of mortgage applications and originations. HMDA data include the lender name, loan amount, property type, loan purpose, and the applicant's income, gender, and race. Location information is limited to the census tract of the house.

Zillow ZTRAX: Zillow produces the Zillow Transaction and Assessment Dataset (ZTRAX), which includes historical tax assessment records as well as information on home sales and loan records from deeds stored in county clerk offices. Each housing transaction includes the date, sales price, mortgage loan amount, mortgage lender's name, and the precise location of the property (address, latitude, and longitude). The tax assessor data includes basic characteristics for each parcel, including assessed land values, total assessed values, and square footage. The transactions and assessor data within ZTRAX can be merged together with a unique parcel-level identifier created by Zillow.

Flood Insurance Policies and Claims: I obtained data directly from FEMA on the universe of flood insurance policies and claims through a Freedom of Information Act request. Location information in both the policies and claims datasets is limited to the zip code and flood zone classification.[9]

Construction Costs: I obtain data on construction costs from R.S. Means, a consulting company and data provider for the construction industry. The dataset includes estimates of annual construction costs at the 3-digit zipcode level, which can vary by the size of the home and other characteristics such as whether the home has a basement. I use

[9] Some of this data is now available publicly through FEMA's OpenFEMA API, but my data includes some variables which are not in the public dataset, such building value assessments as property damage assessments.

construction costs for a 1000 square foot, average quality home without a basement. Because not all 3-digit zipcodes are available, I average across locations to obtain a state-level measure of annual construction costs.

First Street Foundation Flood IQ Model: I obtain parcel-level estimates of flood probabilities from the First Street Foundation Flood IQ dataset as an independent assessment of flood risk that complements my flood maps from FEMA.

Merge Procedure: I merge both the McDash and HMDA datasets with Zillow ZTRAX to obtain the latitude and longitude coordinates of each property, as well as other characteristics of the properties which secure the mortgages. I use a fuzzy merge on zip code, origination year, loan amount, and lender name to execute this, which is based on similar algorithms from the literature (e.g. Bayer et al. (2016)). Appendix Table A.1 provides summary statistics for the analysis sample and the full HMDA, McDash, and Ztrax samples, and shows that the final estimation sample appears broadly representative of the input data sets, even for the variables that were not used in the merge, such as credit scores, debt-to-income ratios, and interest rates. Additional datasets are then merged in at the property level, the zip level, or the state level, depending on the level of granularity.

2.3 Descriptive Evidence

Having assembled this loan-level dataset on flood risk exposures, I now discuss some descriptive facts about mortgages in and out of flood zones.

Panel A of Table 1 shows summary statistics on mortgage characteristics at the loan level. Approximately 20% of the sample of mortgages are located in flood zones. A few notable patterns emerge when comparing mortgage characteristics by flood zone status. Mortgages in flood zones have lower LTV ratios (89.3% versus 86.6%), slightly higher interest rates (4.19% versus 4.2%), and lower delinquency rates (3.79% versus 3.07%) than borrowers outside flood zones. Flood zones also have a different composition of borrowers. On average, borrowers in flood zones have higher credit scores (721 versus 728) and higher incomes ($76,000 versus $100,000). We show additional summary statistics in Appendix Table A.2.

These patterns can also be seen in the histograms. Figure 2 shows the distribution of mortgage terms in and out of flood zones. The distribution for interest rates looks very similar in and out of flood zones (Figure 2a). However, the distribution for LTV ratios differs significantly (Figure 2b), with a much lower fraction of loans with high LTVs in flood zones (LTV $>$ 95%).

One may wonder whether the differences in mortgage terms are entirely driven by the observable differences in borrower composition

and property values documented in Table 1. In Table 2, I show that the broader patterns for LTV ratios (Column 1), interest rates (Column 2) and delinquencies (Column 3) hold even after including zip code - year fixed effects and including borrower-level controls for credit score, income, and property values. With no controls or fixed effects, LTV ratios are approximately 2.7 percentage points lower in flood zones. Including zip code-year fixed effects lowers the coefficient to approximately 1.9 percentage points, showing that much of the variation in LTV ratios is across zip codes. Controlling for income, property value, and credit score further lowers the coefficient to approximately 87 basis points, showing that much of the reduction of LTVs in flood zones can be explained by observable differences in composition, but not all of it. With the fixed effects and controls, LTVs are 83 basis points lower in flood zones. All standard errors are clustered at the county level. These results show that observable differences in borrower composition do not entirely explain the difference in observed LTV.

Column (2) shows that interest rates are only 1 basis point higher in flood zones, even after including the full set of controls and fixed effects. Column (3) shows that delinquency rates are on average 72 basis points lower in flood zones when there are no controls or fixed effects; however, after including the controls and fixed effects, delinquency rates are equal, with standard errors indicating this is a precisely estimated zero. These patterns are consistent with the three hypotheses outlined in Section 1.

I discuss additional descriptive evidence on how other mortgage terms vary by flood zone status in detail in Appendix Section E.1. These results highlight a few key points that help contextualize the LTV and interest rate differentials. First, I show evidence of lender credit rationing on the extensive margin as well, with denial rates also being correlated with flood zone exposure (Appendix Table A.3). Second, I run a pricing regression to show that the interest rate adjustment observed in flood zones makes sense given how the composition of borrowers change (Appendix Table A.4). Third, I consider the natural concern about measurement when certain borrowers may use discount points to obtain a lower interest rate, as is often the case in other contexts ((Hino and Burke, 2021)). I use extended data to verify that the interest rate results are not offset by significant differences in the usage of discount points by borrowers (Appendix Table A.5). Fourth, it is natural to wonder how flood zone status interacts with whether lenders choose to retain mortgages on their balance sheet rather then selling or securitizing them. I show that the LTV reductions are driven by the set of mortgages that are retained on lender balance sheets and not sold to the GSEs. There is no statistically significant reduction of LTVs in flood zones for mortgages sold to Fannie Mae or Freddie Mac in the same calendar year (Appendix Figure A.3).

Limitations of Descriptive Evidence: Taken together, the descriptive evidence shows that LTV ratios are lower in flood zones, interest rates are equal, and that delinquency rates are the same (with a precise zero). This implies that lenders bear less exposure to housing collateral in flood zones, with lending in flood zones not observably riskier than lending outside of flood zones.

What the descriptive facts do not explain is why LTV ratios are lower in flood zones. There are three primary explanations: a supply-side one, and two demand-side ones. The supply-side explanation, consistent with the conceptual framework in Section 1, is that lenders require higher down payments in flood zones to manage their exposure to flood risk. Higher down payments are effective because they lower expected losses by both lowering the probability of negative equity as well as the loss given default. On the other hand, there could also be a demand-side explanation driven by unobserved amenities. In this channel, wealthier individuals who tend to have smaller loans also prefer to live in flood zones. This can occur because flood zones have unobserved amenities, such as water views and beach front access, which disproportionately attract wealthy people. The third explanation is advantageous selection on flood risk itself – that heightened flood risk can directly end up selecting for borrowers with a higher capacity to self-insure. This could occur if, for example, lower wealth borrowers choose to not live in flood zones if they know they cannot self-insure.

Addressing these demand-side explanations requires variation in flood risk that is exogenous to other confounders, like unobserved amenities or unobserved wealth. In Sections 3 and 4, I consider two sources of variation in flood risk that are plausibly exogenous to unobserved variables like amenities or wealth. The goal of these empirical strategies is to identify whether credit supply is reduced in flood zones.

3. Effect of the Flood Insurance Coverage Limit

In this section, I utilize the fact that government flood insurance contracts only cover up to $250,000 in damages to identify the causal effect of flood risk on mortgage terms. The key intuition is that smaller homes can be completely insured, whereas larger homes are only partially insurable and therefore expose lenders to residual flood risk. I show that LTVs are insensitive to flood zome status when the home can be completely insured, and are sensitive to flood zone status when homes can only be partially insured. Moreover, the relationship between flood zone status and LTVs changes at the flood insurance coverage limit.

3.1 Empirical Strategy

As discussed in Section 2, mortgage borrowers in flood zones are required to purchase flood insurance, and FEMA's NFIP insurance program dominates the market. The NFIP only provides up to $250,000 in coverage, and there are limited private options for excess flood coverage in Florida's highest risk flood zones. This leaves a large segment of homeowners in flood zones under-insured. I will exploit this setting to consider how access to full insurance changes mortgage contracts in flood zones.

Banks are not exposed to flood risk when homes are fully insurable because every dollar of flood damage is offset by a dollar of insurance payments and flood insurance is mandatory.[10] Therefore, banks do not need to adjust mortgage terms in flood zones when homes can be fully insured. In contrast, when homes cannot be fully insured, banks retain exposure to flood risk because flood damages can exceed the flood insurance cap and lead to borrower default and damaged collateral. Therefore, banks only have an incentive to ration credit in flood zones for under-insured homes. I formalize this point in Appendix Section B.2.

This intuition leads to the following specification:

$$Y_{izt} = \delta_{zt} + \beta_1 FZone_{it} + \beta_2 CapBinds_{it} + \beta_3 (FZone_{it} \times CapBinds_{it}) + \gamma' X_{it} + \varepsilon_{izt} \tag{1}$$

for a mortgage i originated in year t in a zip code z. The dependent variable Y_{izt} is either the LTV ratio at origination, the interest rate at origination, or an indicator for whether the mortgage becomes delinquent by more than 30 days within the first three years of origination. The indicator $FZone_{it}$ equals one when the property is located in a FEMA-defined flood zone, and zero otherwise. The indicator $CapBinds_{it}$ equals one when the replacement cost of a home exceeds the flood insurance coverage limit, and 0 otherwise. The specification includes zip-year fixed effects in δ_{zt}. These are important since they absorb any time-varying local shocks at the zip code level which could also influence LTV ratios. In some specifications, I also include a rich set of mortgage and borrower controls in X_{it} to separate the treatment effect of being in a flood zone from any effects due to differences in the composition of flood zones. Borrower characteristics include the borrower's credit score and annual income at origination. Mortgage characteristics include the property value, maturity, debt-to-income ratio, and combined LTV ratio for other liens on the property. I also include indicator variables for mortgage type, such as whether it is a first mortgage, second home, low grade mortgage, full document mortgage,

[10] Even if insured homeowners in flood zones choose to default after floods, lenders are still entitled to receive insurance proceeds in foreclosure (Hoberock and Griebel, 2018; Issler et al., 2019).

jumbo loan, FHA/VA backed, or an adjustable rate mortgage. Finally, I control for flood insurance take-up rates which vary at the flood zone - zip code - year level. Standard errors are clustered at the county level.

The key parameters of interest are β_1 and β_3. For homes that can be fully insured ($CapBinds_{it} = 0$), LTV ratios in and out of flood zones should be the same because flood risk is fully insurable and therefore banks do not need to ration credit. This suggests that the estimated coefficient $\beta_1 = 0$. For homes that cannot be fully insured ($CapBinds_{it} = 1$), LTV ratios should be lower in flood zones, because banks retain exposure to flood risk. This suggests that the estimated coefficient $\beta_3 < 0$.

Measuring Whether the Insurance Cap Binds: To determine whether the insurance coverage limit can bind, I consider whether a home's replacement costs at origination exceeds the $250,000 flood insurance cap.

Insurance companies estimate replacement costs to help households determine how much insurance coverage they need. The point of this measure is to capture the cost of rebuilding the exact same house in the event of a total loss disaster. If the home is completely destroyed, the homeowner pays the difference between the cost of rebuilding the home and insurance claim pay outs. If coverage is below replacement costs, then households will be underinsured, with some repair costs not covered by insurance. Replacement costs depend on a variety of factors, including local construction costs, square footage, year built, the quality of materials used to build the home, and other home features. I develop a proxy for this measure by multiplying construction costs for an average quality home (dollar per square foot) and the size of the home (square feet). I obtain annual construction costs for Florida from the R.S. Means company, and I obtain the building size from Zillow ZTRAX's assessment dataset (see Section 4 for more details). This methodology follows the procedure described by insurers for estimating replacement costs.[11]

A reader may naturally wonder why I do not use house prices as a rough proxy of replacement costs, instead of introducing this new measure. One may think that replacement costs and property values have a monotonic relationship, with high market value homes having high replacement costs. However, in reality the two measures can be quite different; for example, land values and amenities are a component

[11] For example, the insurance trade group Insurance Information Institute (III) writes: "In every case, you'll want the limits on your policy to be high enough to cover the cost of rebuilding your home. The price you paid for your home—or the current market price—may be more or less than the cost to rebuild [...] For a quick estimate of the amount of insurance you need, multiply the total square footage of your home by local, per-square-foot building costs. (Note that the land is not factored into rebuilding estimates.)".See Insurance Information Institute, "How Much Homeowners Insurance Do You Need", available at: https://www.iii.org/article/how-much-homeowners-insurance-do-you-need.

of property values but do not impact replacement costs. Appendix Section D discusses in more detail why it would be incorrect to use house prices to determine whether the flood insurance cap binds.

One might also think that \$250,000 is sufficient coverage, and that floods do not cause high enough levels of damage for the insurance coverage limit to bind or be relevant to banks. In Figure A.2a, I plot the distribution of flood insurance claims for each replacement cost value estimated by FEMA. While large floods are rare, flood damages exceed the insurance for large homes roughly 10 percent of the time, showing that uninsurable damages are a relevant consideration for lenders.

Next, I estimate whether the effect of flood zone on LTV ratios is larger for homes that have a larger portion of the home uninsured. To do so, I replace the $CapBinds_{it}$ indicator in Equation 1 with an indicator for replacement cost categories $RCost_{k,it}$. I create these k categories using replacement cost increments of size \$25,000; for example, homes with an estimated replacement cost of \$100,000 and \$120,000 will be included in the same replacement cost category. I then estimate the following specification:

$$Y_{izt} = \alpha_{zt} + \delta FZone_{it} + \sum_{k \neq 1} \theta_k RCost_{k,i} + \sum_{k \neq 1} \phi_k (FZone_{it} \times RCost_{k,i}) + \gamma' X_{it} + \varepsilon_{izt} \tag{2}$$

for a mortgage i originated in year t in a zip code z. As earlier, $FZone_{it}$ is an indicator variable which equals one for homes that are located in a FEMA-defined flood zone and zero otherwise. I include zip-year fixed effects α_{zt} and the same set of loan-level control variables in X_{it} that I used in Equation 1. The key parameters of interest here are the ϕ_k coefficients, which show the average reduction of LTV ratios in flood zones for that replacement cost category relative to the omitted category of homes with replacement costs less than \$75,000 (category $k=1$). For homes with replacement costs less than \$250,000, I expect $\phi_k = 0$. For homes with replacement costs above \$250,000, I expect $\phi_k < 0$. Furthermore, I expect the effect of being uninsured to increase with how much of the home remains uninsured, with $\phi_{k+1} < \phi_k < 0$. Standard errors are clustered at the county level.

Identifying Assumptions: The key omitted variable in this setting is unobserved amenities, because wealthier people may disproportionately move to flood zones because of amenities, and such borrowers tend to choose mortgages with higher down payments. As a result, any trend in LTV ratios may be driven by borrower preferences due to unobserved amenities rather than flood risk. Here I try to deal with this endogeneity concern more directly by exploiting variation in flood risk exposure that is plausibly exogenous to unobserved amenities.

Equation 2 is a cross-sectional difference-in-differences design. The key identifying assumption is that, conditional on controls, there is no discontinuous change in unobserved amenities of flood zones around the flood insurance coverage limit. This approach accommodates a correlation between unobserved amenities and flood zone, as well as unobserved amenities and replacement costs. The functional form assumption is that these biases are additive, meaning that the expected value of LTV ratios for homes with no flood risk exposure can be written as the sum of a flood zone fixed effect and a replacement cost fixed effect. Thus, changes in mortgage terms around the flood insurance coverage limit are unlikely to be attributable to changing amenities.

Like most empirical designs, this exogeneity assumption cannot be tested directly because amenities are unobserved. However, it can be partially evaluated by observing whether the relationship between mortgage terms and replacement costs for homes below the insurance cap are similar in and out of flood zones. This should be the case because homes in flood zones below the insurance cap are fully insured against flood risk. In other words, I can check whether the coefficients for ϕ_k in Equation 2 equal to zero for homes with replacement costs less than 250,000. This would also support the identifying assumption that amenities in flood zones do not change with replacement costs.

3.2 Results

Table 3 reports the estimates of Equation 1 for LTV ratios (Panel A), interest rates (Panel B), and delinquency (Panel C). Column (1) considers the specification without any fixed effects or controls. The subsequent columns add in the fixed effects and control variables sequentially, as indicated in the bottom of the table. The "restricted" controls exclude borrower income and credit score, and the "full" controls include these. Column 1 of Panel A reveals that on average, with no controls or fixed effects, LTV ratios are lower in flood zones for all replacement cost groups. This is consistent with flood zones having amenities that may attract richer borrowers, as well as credit rationing by lenders. These patterns persist in Column (2) after including zip-year fixed effects, although the magnitudes shrink. After including control variables in Column (3), there is no statistically significant (or economically significant) effect of flood zone on LTV ratios when homes can be completely insured. However, for homes that are under-insured, being in a flood zone leads to a 0.85 percentage point decrease. The results are similar in Column (4) after including income and credit score controls. The LTV reduction for under-insured borrowers in flood zones persists despite controlling for property values, borrower incomes, and credit scores, suggesting that borrower characteristics do not entirely explain the differences in observed LTVs. This suggests that the capacity

to self-insure does not entirely drive the LTV results, to the extent that this is captured by these controls. Taken together, this table shows the following result: two otherwise observably-equivalent borrowers that differ only by flood zone have the same LTV if their replacement cost is below $250,000, but have different LTVs if their replacement cost is above $250,000.

For interest rates in Panel B, Column (1) reveals that on average, borrowers with high replacement costs in flood zones have slightly lower interest rates (by 3 basis points), but this result becomes a precisely estimated zero after including fixed-effects and controls. In Column (4), there is no statistically significant difference in interest rates for either under-insured or fully-insured borrowers in flood zones. Lastly, Panel C focuses on the patterns of delinquency. In Column (1), there is a significant decline in delinquency for borrowers in flood zones that are under-insured by 0.77 percentage points. Delinquency rates are also significantly lower for households with high replacement costs outside flood zones (by 2.4 percentage points). High replacement cost borrowers in flood zones have slightly higher delinquency rates, though this is not statistically significant. In the specification with the full set of fixed effects and controls in Column (4), it is clear that much of this difference in delinquency is due to borrower composition – after controlling for these characteristics, delinquency in and out of flood zones are nearly identical for both the fully-insured and under-insured borrowers. The three panels together show that LTV ratios are significantly lower for two observably-identical borrowers in flood zones that are under-insured, while interest rates and delinquency are virtually the same.

I then consider this same analysis by looking at borrower income and credit scores as outcome variables for the regression in Equation 1. If there truly is lender rationing, then one might expect there to be distributional effects in terms of who is able to obtain a mortgage. This is because higher down payment requirements could screen out more liquidity constrained borrowers from purchasing high replacement cost homes in flood zones. This hypothesis implies that incomes and credit scores should be higher for under-insured borrowers in flood zones; Appendix Table A.6 confirms this intuition. Columns (1) and (3) exclude zip-year fixed effects, and Columns (2) and (4) include them. The results show that incomes are far higher in flood zones for both the under-insurable and fully insurable groups. Credit scores are only higher in flood zones for the under-insured group; credit scores are 2 points lower for the fully-insured group in flood zones.

To test whether the results on mortgage terms are driven by the flood insurance coverage limit directly, I consider how the effect of flood zone status on mortgages varies by the replacement cost of the home. Figure 3 plots the ϕ_k coefficients obtained by estimating Equation 2 for

LTV ratios (panel A), interest rates (panel B), and delinquency (panel C). Figure 3 includes the specification with zip-year fixed effects and restricted controls (i.e. without controls for income and credit score). The plots reveal that there is no average difference in LTV ratios in and out of flood zones when replacement costs are lower than the $250,000 flood insurance coverage limit. However, once replacement costs pass the insurance coverage limit, LTV ratios are significantly lower in flood zones. Furthermore, the magnitude of the coefficient increases with how much of the home remains uninsured. Homes with a replacement cost of $300,000 in flood zones have LTV ratios that are 1 percentage point lower than homes in the same group outside of flood zones. For homes with a replacement cost of $425,000, LTV ratios are more than 2.5 percentage points lower in flood zones. The figure looks qualitatively and quantitatively similar in Appendix Figure A.4, which adds controls for income and credit score.

Another way to think about this exercise is to use down payments directly as the outcome variable rather than LTV ratios, where this is calculated simply as the difference between the property value and the loan amount at origination. Appendix Figure A.5 confirms that down payments are no different in and out of flood zones for fully insured homes, but begin to increase in flood zones as homes become under-insured. The magnitudes are intuitive – for example, a home which is under-insured by $100,000 seems to have an increased down payment of $5,000, holding fixed the property value. In other words, every uninsured dollar requires an additional down payment of roughly 5 cents.[12]

Securitization: A natural question concerns how the flood insurance cap may interact with lender incentives to screen given the ability to sell mortgages to the government-sponsored enterprises. In Appendix Section E.2, I consider a number of alternate tests to consider how to think about the results in light of the outsized role of the government-sponsored enterprises in mortgage markets and the effect on lender screening incentives (Ouazad and Kahn, 2021; Sastry, Sen, and Tenekedjievia, 2023). These tests show that my estimates of the effects of the flood insurance cap are not driven by confounding effects stemming from GSE securitization. This section shows that the LTV reductions are concentrated in mortgages retained on lender balance sheets (Appendix Table A.7); that there is limited of strategic securitization behavior around the flood insurance cap (Appendix Figure A.6); and that the replacement cost limit and conforming loan limit do not match up (Appendix Figure A.7).

[12] This also roughly matches the probability of at least one flood over the life of the loan, assuming a 1% annual flood probability and a 5 year life of a loan.

4. Effects of Updated Flood Maps on Mortgages

In this section, I consider the effects of updated FEMA flood maps that expand the boundaries of high risk flood zones. I use this second source of variation to explore whether mortgage markets dynamically respond to new information about risk. This is particularly important given that climate change is a dynamic process that will likely require continual adjustments over time. I show that mortgage terms change after the release of the updated maps with reduced LTV ratios for under-insured high risk borrowers, and a compositional shift to richer and higher credit score borrowers.

4.1 Background

I consider a natural experiment in which flood risk assessments are updated while other aspects of an area remain fixed. This creates variation in flood risk assessments that are independent of unobserved amenities in a flood zone, the key omitted variable of concern. Obtaining random variation in fundamental flood risk is challenging because flood risk changes extremely slowly over time. In this section, I follow the literature and consider variation in information about flood risk (Giglio, Kelly, and Stroebel, 2021). I look particularly at updated flood maps from FEMA, which release new information on risk and also correlate with actual changes in fundamental risk due to the construction or destruction of dams and levees. As discussed in Appendix Section C.1, FEMA produces updated flood maps that include new data on elevation on land erosion which are essential inputs for modeling flood risk and used by private sector modelers (National Research Council, 2007).

The release of an updated map has two effects. First, banks must update their compliance systems and notify any mortgage borrowers in newly mapped flood zones to buy flood insurance; this applies to both existing and new borrowers. In Appendix Table A.10, I verify that new borrowers in flood zones do also purchase flood insurance, showing evidence of lender compliance with the flood insurance mandatory purchase requirement.[13] Second, these maps arguably provide new information that change flood risk assessments without changing other features of an area that can induce borrower selection, such as coastal amenities. These new flood maps are extremely time-consuming and costly to produce, and they often produce new data that are subsequently used by the private sector to model flood risk. I argue that banks utilize the new information released by FEMA, and in the rest of this section, I explore the effect of these updated maps on mortgage terms.

[13] This is consistent with flood insurance take up evidence from Mulder, 2022.

4.2 Empirical Specification

New flood maps do not always lead to heightened perceptions of risk. Some maps expand the boundaries of flood zones in a zip code while others may keep boundaries the same or even contract them. For example, flood boundaries may contract because communities construct levees and dams to manage water flow. On the other hand, flood boundaries may expand if new development raises the surface elevation of water. To understand in which direction the maps change risk assessments, I classify each zip code by whether the new maps expand flood zone boundaries or contract them. I make this determination by comparing the share of homes in a flood zone under the old map to the share of homes in a flood zone under the new map for a given zip code. I also check to make sure that the updated map is actually the expansion of an existing flood zone rather than a shift in the location of the flood zone by checking that homes in a flood zone under the old map are also in a flood zone under the new map. On average, for new maps that expand flood zone boundaries in a zip code, the zipcode share of homes in a flood zone increased by 30% (or by 4 percentage points).

To implement the difference-in-differences design, I restrict the sample and define treatment as follows. I first exclude any counties that are re-mapped multiple times in sample. Second, if a county c receives a new map, I classify a zip code z in that county as "treated" if the new map expands that zip code's flood zone boundaries. That is, I exclude zip codes in the county that receive a new map but have unchanged or contracted boundaries. I do so to isolate areas that lead to a *heightened* assessment of flood risk, which would naturally occur when flood boundaries expand. I consider these alternate zip codes as a placebo group in a robustness check. Third, because my sample spans 2010-2016, I limit to counties that are remapped in either 2012, 2013, or 2014 to ensure I have enough years before and after the remappings to estimate pre-trends and treatment effects. Appendix Figure A.8 shows which counties were remapped in which year. In the main specification, I exclude counties that do not receive an updated map between 2005 - 2016, but show in a robustness test that the results are not affected by including this "never-treated" group.

I estimate the following specification at the mortgage level:

$$Y_{i,c(z),t} = \alpha_{c(z)} + \delta_t + \sum_{h \neq -1} \beta_h \mathbb{I}\{E^{Expanded}_{i,c(z),t} = h\} + \gamma' X_{it} + \varepsilon_{i,c(z),t} \tag{3}$$

The dependent variable $Y_{i,c(z),t}$ is the mortgage's LTV ratio at origination, interest rate, and other outcomes of interest. For each mortgage i that is originated at time t, I identify whether its county c receives an updated map, the year of the updated map $\tau_{c(z)}$, and whether it is in a treated zipcode z that has an expanded flood zone.

I construct the event-time variable $E^{Expanded}_{i,c(z),t} = t - \tau_{c(z)}$ which reflects the mortgage's origination year relative to the release of the county's updated map. The variable is defined for zipcodes where updated flood maps expanded flood zone boundaries. I include year fixed effects δ_t and county fixed effects $\alpha_{c(z)}$ which control for any unobserved year or county shocks. In some specifications, I also include loan-level controls, represented by X_{it}. Treatment occurs at the county level since all communities in a county receive a new flood map at the same time. I therefore cluster standard errors at the county level (Bertrand, Duflo, and Mullainathan, 2004).

The key parameters of interest in Equation 3 are the β_h coefficients on the event-time indicators which estimate the outcome at a given event-time relative to the omitted category $h = -1$, the year prior to the updated map.

I also consider the pooled specification, replacing the event-time dummy with the indicator $Post^{Exp}_{c(z)t}$ that equals 1 after an area receives a map update that expands flood boundaries and equals 0 beforehand. As earlier, the variable is not defined for zipcodes which receive an updated floodmap that contracts flood zone boundaries.

Identifying Assumptions: A key assumption of the staggered difference-in-differences design is that LTV ratios among the treated and control groups would have evolved in parallel in the absence of any re-mappings. Under the common trends assumption, for all dependent variables, coefficients on the event-time indicators before the remappings should be zero ($\beta_{-3}, \beta_{-2} = 0$).

This empirical strategy uses variation in both the location and timing of the release of updated maps. Therefore another key assumption of this approach is that the timing of map updates is uncorrelated with other determinants of LTV ratios. To help assess the validity of this assumption, Table 4 shows descriptive statistics on mortgages and flood insurance by remapping year. There are no systematic differences in most socio-demographic characteristics across zipcodes based on when they receive a new map. I also check FEMA's publications on how they prioritize new map updates to ensure their decision rule does not depend on variables that could be endogenous to LTV ratios, such as income or population growth. FEMA chooses to update maps if the prior map is very old or if there is evidence that the prior map is inaccurate, such as large losses or high insurance takeup outside of flood zone boundaries (National Research Council, 2007).

4.3 Main Results

Table 5 shows the main difference-in-differences estimation results and reports the β_h coefficients from Equation 3 for LTV ratios (Columns 1-4), log income (Column 5) and credit score (Column 6). Column (1)

shows the results with no controls; Columns (2) through (4) sequentially include different controls for borrower characteristics as well as flood insurance take-up. Consistent with Hypothesis 1, Column (1) shows that LTV ratios decline in the years following the introduction of the updated map. In the first year after the remapping, LTV ratios decline on average by 0.48 percentage points, though this result is not statistically significant. The reduction is larger in the second year, closer to 1.5 percentage points. By the third year, LTV ratios are more than 2 percentage points lower. This suggests that after the remappings expand flood zone boundaries in a zip code, borrowers on average receive loans with lower LTV ratios. Importantly, the coefficient estimates from the pre-period support the parallel trends assumption. Columns 2-4 show that these LTV results are robust to including controls for borrower income, borrower credit scores, and flood insurance takeup rates. The results show that two borrowers with the same income and credit scores would still obtain different mortgage contracts after the release of the updated maps. This shows that changing borrower characteristics do not entirely explain the differences in observed LTVs, suggesting that the capacity to self-insure does not entirely drive the LTV results.

Borrower Composition: The next results in Table 5 show the effects of the remappings on borrower composition. Column (5) reports the results using log income as an outcome variable, defined as annual applicant income at mortgage origination reported in the HMDA data. In the first year after the remapping, incomes significantly increase by 5%. In the second year after the remapping, incomes increase by almost 10%, and this number increases further to 15% in the third year. Column (6) shows similar results for credit scores. Credit scores increase by 3 points in the first year after the remapping. Two years after the remapping they are 6 points higher, and three years after the remappings they are 8 points higher. There are also no visible pre-trends in either variable.

Other Mortgage Terms: Table 6 shows the pooled results for other mortgage terms, showing they do not significantly change after the remappings. Column (1) in Table 6 shows the estimation results for interest rates. Consistent with Hypothesis 2, interest rates on average slightly increase by 6 basis points, though this is not statistically significant. Column (3) of Table 6 shows that debt-to-income ratios on average reduce in the post-remapping period, though not significantly so. Column (4) shows that average mortgage maturity reduces by less than one month after the remappings, and this result is also not significant. The results point to LTV ratios as the key margin of adjustment.

Delinquency: Column (2) of Table 6 shows the results for delinquency rates. Delinquency rates reduce on average by nearly

1 percentage point after the mappings. Taken together, after the remapping, mortgages have both lower LTV ratios and a lower likelihood that the mortgage becomes delinquent within the first three years of origination.

Flood Insurance: One may wonder what role might be played by increasing flood insurance purchases, such as if the requirement to buy flood insurance may change borrower demand for high-LTV or low-LTV loans. While it is not directly possible to separate the independent effects of insurance and updated information in this empirical approach, the following results suggest that insurance take-up plays less of a role in driving the LTV results from the updated maps. First, Table 5 shows that the LTV results are robust to controlling directly for flood insurance purchases. Second, Appendix Table A.11 explores the correlation between flood insurance take-up rates and LTV ratios in and out of flood zones. This table shows that the relationship is virtually flat, suggesting limited potential crowd-out between flood insurance take-up and LTVs.

House Prices: An important question is what happens to property values, given that LTV is driven by the joint movement of both property values and loan size. One may think that the effect of higher down payment requirements by lenders could be offset by declines in house prices, if flood risk gets directly capitalized into house prices or if the decline in credit supply itself leads to house price declines.[14] To examine this question, I examine the effect of the remappings on capitalization using dollar-per-square-foot as an outcome variable, calculated by dividing property values by the total size of the property (measured in square feet). Column (5) of Table 6 shows the results. On average property values increase by $7 per square foot after the remapping, and the result is not statistically significant. This insignificant result carries the following interpretation: house prices do not fall by enough to offset the higher down payment being demanded by the bank, meaning that constrained borrowers who would like to seek out cheaper housing today as a way to relieve constraints are limited in their ability to do so.

While perhaps surprising, the insignificant result on house prices is consistent with the largely mixed results in the literature.[15] The closest paper in spirit to this one is Hino and Burke (2021), who look nationally and for a longer time-series, but also find that being zoned into a FEMA

[14] The latter would occur in models of durable goods such as Rampini (2019), because down payment-constrained borrowers would react by seeking out lower quality assets (i.e. less durable assets), that are cheaper today but likely to appreciate in price tomorrow. In this case, declining credit supply would also lead to a decline in house prices.

[15] Relevant papers include: Baldauf, Garlappi, and Yannelis (2020); Bernstein, Gustafson, and Lewis (2019); Gibson and Mullins (2020); Giglio et al. (2021); Keenan, Hill, and Gumber (2018); Murfin and Spiegel (2020)

flood zone does not lead to statistically significant changes in house prices. I further discuss the relationship between my results and the theory on what might expect for house prices in Appendix Section E.3. On the one hand, rising flood risk may lower housing demand, pushing house prices down; on the other hand, flood risk changes financial constraints, housing supply, and housing quality, which would push house prices up. Taken together, the predictions for rising flood assessments on house prices are mixed.

4.4 Effect of Flood Insurance Cap

I now consider a triple difference-in-differences design to examine whether LTV reductions are higher for the group which would be under-insured against flood risk due to the $250,000 cap on flood insurance, as implied by the results in section 4. I specifically consider the following interacted specification:

$$Y_{i,c(z),t} = \beta_1 Post^{Exp}_{c(z)t} + \beta_2 (Post^{Exp}_{c(z)t} \times CapBinds_i) + \alpha_{c(z)} + \delta_t + \gamma' X_i + \varepsilon_{i,c(z),t} \tag{4}$$

This specification looks at mortgage terms both before and after the release of the updated maps, and splits the effect of the remapping by whether the household will be fully insured (coefficient estimate β_1) or will be underinsured (coefficient estimate β_2). All specifications control linearly for the main effect of replacement costs (in X_i). In some specifications, I also include loan-level controls for borrower income and credit scores (also in X_i).

The results of this specification are reported in Table 7. The two dependent variables are the mortgages' LTV ratio at origination (Columns 1-2), and interest rate at origination (Columns 3-4). Columns 2 and 4 include loan-level controls for income and credit score, whereas Columns 1 and 3 do not. The results indicate that, indeed, the LTV reduction following the remapping events is driven by the higher replacement cost groups. Those households have on average a 0.81 percentage point LTV reduction; this holds even after holding fixed borrower characteristics, though the coefficient shrinks to 0.66 percentage points. This result implies that the under-insured households do have significantly lower LTVs, consistent with the results in Table 3.

For interest rates, Columns 3 and 4 show that on average interest rates for the under-insured groups decrease by 3 basis points following the remappings, and this number shrinks slightly to 2 basis points after including the loan level controls. This interest rate result suggests that there may be indeed some demand-side changes that partly explain the LTV results, since some under-insured borrowers get lower LTVs and slightly lower interest rates. These are, perhaps, individuals with greater

capacity to self insure who trade off the costs of self-protection with the benefit of lower interest rates.

It may be surprising that LTVs also rise for fully insurable properties in Table 7. A natural way to think about this result is the sorting that is induced by a change in lender menus after the remapping. Consider the marginal liquidity constrained buyer that is choosing between a fully insurable home and a partially insurable home. Prior to the updated flood map, such buyers might have chosen the higher replacement cost home with a higher LTV; however, after the updated flood map, such buyers cannot get the same home for the high LTV due to lender rationing, and end up choosing lower replacement cost homes, which in turn require putting less money down. This result can be thought of as reflecting heterogeneity in the causal effect of changing LTV menus brought by updated flood maps. Importantly, these results do not violate the key identifying assumptions of this design: (1) that the timing of map updates are plausibly exogenous and that (2) there are parallel trends in LTVs between remapped and not-yet-remapped zip codes prior to the map updates. This is because, in this empirical design, treatment is assigned at the zip code level, with all borrowers in a remapped zip code being treated.

Heterogeneity: I further explore these results by considering heterogeneous effects by borrower risk and insurability against flood risk. In Figure 4, Panel A, I split the sample into two groups: a low borrower risk group defined by both high house prices (above $200K) and high credit scores (above 700), and a high borrower risk group defined by either low house prices (below $200K), low credit scores (below 700), or both. Panel A shows that there is a statistically significant LTV reduction for the high risk borrowers, but not for the lower risk borrowers. This may be surprising since one might think that the high risk group is likely to be fully insurable, because they have low house prices. However in reality there is significant heterogeneity in replacement costs for these borrowers. In Panel B, I focus on the sample of high risk borrowers, and then consider three replacement cost groups: those that are just below the cap and fully insurable (between $200K-$250K), those that are just above the cap and under-insured (between $250K-$350K), and those that are highly under-insured (above $350K). As can be seen, the LTV reduction is driven by the high risk borrowers with uninsurable flood risk. Borrowers that are fully insurable see no statistically significant change in LTVs after the remappings. For robustness, In Panel C, I remove properties at the boundary (within $10,000 of the flood insurance cap) to show that the results are not driven by those properties most susceptible to measurement error in replacement cost; as expected, the LTV reduction for properties just above the cap gets even stronger. In Panel D, I subset to the very high

risk borrowers with both low credit scores and low property values. In this group, the LTV rationing becomes even stronger for uninsurable borrowers.

These results help to support a supply-side interpretation of the LTV results, suggesting that lenders respond to the updated flood maps by rationing credit specifically to borrowers with both high credit risk and highly uninsurable flood risk. If lenders believe borrowers have low default risk to begin with, there may be less need to worry about default triggered by uninsurable property damage after a storm. Such borrowers are likely less liquidity constrained, and could be more likely and willing to have financial resources to cover uninsurable damages themselves since default is quite costly. In contrast, higher risk borrowers may have less ability to smooth shocks and could be more dependent on flood insurance payments to stave off default. Flood insurance itself is highly imperfect– among the most well-known issues is the uncertainty and delay surrounding the payments. FEMA cautions that it can take nearly two months before a claim is finalized and paid, and the repair process could itself take months to be completed. Such delays can be particularly costly for the highest risk borrowers that may be more constrained in their ability to smooth disaster shocks, making them more likely to default on their mortgage instead of repaying. Even if lenders eventually receive flood insurance proceeds, it may come with a significant lag, and for under-insured properties the amount may not be enough to offset lender losses. It makes sense that lenders may therefore focus their credit rationing on higher risk borrowers with uninsurable flood risk.

4.5 Robustness

The staggered difference-in-differences design may raise a few possible identification-related concerns. I consider them one-by-one, and offer robustness checks to address them.

Mortgage Denials: To help understand demand and supply-side explanations, I also consider what happens to mortgage denials after the release of the updated flood maps that expand flood zone boundaries. As argued in prior literature (Gete and Reher, 2018), mortgage denials are decided at the discretion of lenders, with increases in denial rates being interpreted as reductions in credit supply. In Appendix E, I discuss the details of this this test, including the challenges of using the denials data and how the test must be implemented. The results of the triple difference-in-differences specification can be found in Table Appendix Table A.12. Column (1) shows that, on average, mortgage application denial probabilities increase after the remappings specifically for the high loan size group. . This suggests that lenders pull back from larger loans after the remapping, consistent with a pull-back from high-LTV

lending. In addition, mortgage denials due specifically to insufficient down payments also increase following the remapping, also highlighting the importance of down payments.

Securitization: As earlier, one may wonder if lender securitization behavior may change following the release of the updated maps. To address this concern, Appendix Figure A.9 shows that lenders are not more likely to originate conforming loans following the release of the updated maps.

Beliefs: One may also wonder about the role played by changes in climate beliefs or individual risk preferences around the release of the updated maps. Appendix Section E.3 provides suggestive evidence that the LTV results are unlikely to be driven by changes in beliefs, measured by political affiliation, or risk preferences, measured by flood insurance insurance deductible choices (Appendix Figure A.10).

Never-treated counties: In the main specification of Equation 3, I exclude counties which do not receive updated flood maps at all, the never-treated sample. I show in a robustness check that the main results are robust to including never-treated counties in the control group, defined as counties which do not receive updated flood maps between 2005-2016. Here, the event-time variable is defined for zipcodes where updated flood maps expanded flood zone boundaries and equals zero for never-treated counties in the control group which are not re-mapped between 2005-2016. Appendix Figure A.11 shows the main results when including these counties as part of the regression. The results for LTV ratios, incomes and credit scores are very similar in magnitude to the results from the main specification. Appendix Table A.9 also shows that other mortgage terms (interest rate, DTI, maturity) are also similar, and that the delinquency result is also robust.

Unobserved County-Year Shocks: One possible identification-related concern is that the results may be driven by unobserved county-year shocks rather than the causal effect of the remapping. To address this concern, I conduct a placebo test by running the same staggered difference-in-differences specification for zipcodes that receive updated flood maps which do not change flood zone boundaries or slightly shrink them (by less than 1 percentage point). The idea behind this robustness check is that if my results were driven by unobserved county-year shocks that are unrelated to flood risk perceptions, then there would be similar effects on mortgage terms in zip codes that receive a new map where boundaries do not meaningfully change. Appendix Figure A.12 plots the results for both the treated group and the placebo group of zip codes that received updated maps but had unchanged boundaries The results are starkly different for the placebo zip codes when compared to the zip codes with expanded boundaries. While the placebo group has a slight declines in LTV ratios following the remappings, it is not statistically

significant and the reduction never exceeds 1 percentage point. There are no significant changes in credit scores or log incomes following the remappings. This suggests that unobserved county-year shocks do not drive the main results.

Staggered Design: The recent econometrics literature suggests that staggered difference-in-differences designs can be biased when treatment effects are heterogeneous by treated cohort and/or over time (Callaway and Sant'Anna, 2021). The literature proposes a variety of approaches to address these concerns. Here, I consider the independent effect of the 2012 re-mappings, which employs a standard difference-in-differences design comparing counties remapped in 2012 to counties that are not remapped. This approach has the disadvantage that the sample size diminishes significantly, but has a straightforward interpretation of the dynamic treatment effects. Appendix Figure A.13 shows that the results for LTV ratios, incomes, and credit scores are consistent with what I find in my staggered setting, helping to alleviate the concerns about biased estimates due to the staggered design.

Effectiveness of Adjustments using evidence from Hurricane Isaac: A natural question is whether there is evidence that the LTV adjustments induced by the updated flood maps actually improve default performance after storms. In Appendix Section E.4, I examine this directly by exploiting the landfall of Hurricane Isaac, and considering the effect of LTV on delinquency. The test is described there in detail, but the broad idea is to examine the effect of the hurricane on default, looking at heterogeneity by both loan-to-value ratio at origination, and by whether the mortgage was originated under an old flood map or a new flood map. The results are shown in Appendix Table A.14. I find that in general households with lower loan-to-value ratios default less after Hurrcane Isaac. Importantly, even after conditioning on this direct relationship between LTV and default, there is an additional effect on delinquency based on whether a household's has a lower LTV because the mortgage is originated after a new flood map. This evidence suggests that the greater equity induced by the change in the flood map matters for mortgage delinquency after the storm.

4.6 Magnitudes

One may wonder whether the observed reduction in LTV ratios is an over- or under-reaction relative to the true delinquency risk that banks face from uninsured flood risk. A back-of-the-envelope calculation suggests that the LTV reduction is correct.

The results show that lenders reduce LTVs by roughly 1 percentage point due to flood risk, after controlling for borrower composition. The key question is whether a roughly 1 percentage point LTV reduction in flood zones effectively improves expected lender recovery enough to

offset the increased risk from lending in flood zones. Lender recovery data are unfortunately unavailable, but under the assumption that lender recovery rates and costs in delinquency are similar by flood zone, then one could examine whether the delinquency reduction from lower LTVs is similar in magnitude to the delinquency risk brought by floods.

The first question is what is the delinquency risk associated with floods? This roughly boils down to obtaining the probability of uninsured flood losses over the life of the mortgage. Homes in a flood zone have a 1% annual flood probability –this is the estimate provided by FEMA, and this also lines up with the annual probability of filing a flood insurance claim. Appendix Figure A.2a shows that, conditional on having a flood, roughly 10% of floods cause damages that exceed the $250,000 insurance coverage limit. The average mortgage duration in my sample is roughly 5 years (this also matches the national average, according to the Urban Institute). Thus, the probability of uninsured flood losses over the average life of a mortgage is 50 basis points.

Now, the question is whether a 1 percentage point LTV reduction can lower delinquencies by 50 basis points. Table 5 shows that LTV ratios decline by 2 percentage point after the remappings. Table 6 shows that delinquency rates decline by almost 1 percentage points after the remappings. Since a 2 percentage point LTV reduction is associated with a 1 percentage point decline in delinquency, this suggests that 1 percentage point LTV reduction would reduce delinquency rates by 50 basis points.[16]

Taken together, this back-of-the-envelope calculation suggests that reduced LTV ratios bring lower delinquency rates, and that the level of adjustment is enough to equalize delinquency rates in and out of flood zones.[17]

4.7 Reconciling the Results from the Two Approaches

In this section, I consider how to qualitatively interpret the results across the two empirical approaches.

LTVs and Delinquency: In the first approach of Section 3, I do what is in spirit a cross-sectional comparison of households with high replacement costs that live in flood zones ("high risk type") to households with high replacement costs that live outside of flood zones ("low risk type"). I find that the high flood risk type households pay

[16] To know whether the adjustment is correct, one would ideally obtain a precisely estimated elasticity of delinquency with respect to LTV at origination.

[17] Whether this is the *optimal* level of adjustment requires a model and will depend on assumptions about the distribution of flood damages, the elasticity of delinquency with respect to LTV ratios at origination, the competitive structure of lending markets, and general equilibrium effects on house prices. Future work will bring more evidence to bear on the correct modeling assumptions and parameter estimates.

higher down payments than the low flood risk type, but both have equal delinquency rates. The fact that delinquencies are the same for both types suggests lending in flood zones becomes no riskier than lending outside of flood zones because the contract structure offsets the incentives to strategically default. In the second analysis of Section 4, I also find that LTVs decline after the release of the updated maps, but that delinquencies decline as well. This second analysis reveals that prior to the updated maps, there were likely high risk type borrowers who got higher LTVs than they would have received if the bank knew their true flood zone. After banks obtain new information about their true flood risk, such borrowers end up being forced to put more equity upfront– but having more equity reduces their incentive to strategically default. As a result, delinquency rates decline after the remapping. This result supports the conjecture for the first analysis, that the "counterfactual" delinquency rate of high risk type borrowers would be higher if they did not get rationed.

Interest Rates: In addition, across the three sections of the paper, I find significant adjustments of LTV ratios without significant changes in interest rates. Given the hypothesis in Section 1, it may be surprising that empirically I obtain evidence that LTVs seem to adjust but interest rates do not. There are a number of reasons for why this might be the case. Theoretically, mortgage markets are known to feature both moral hazard and adverse selection, which changes the predictions for how interest rates should adjust relative to the model of just moral hazard in Section 1 (e.g. Einav, Jenkins, and Levin, 2012; Gupta and Hansman, 2022).[18] That said, the combination of both frictions further favors the use of down payments, since higher down payments unambiguously lowers both moral hazard incentives and induces advantageous selection.[19]

Furthermore, there are other institutional features of mortgage markets and flood risk that suggest interest rate adjustments may be less effective in this context. First, Florida and other high climate risk states are vulnerable to large climate disasters every year– if a large disaster occurs early on in the mortgage, lenders do not gain much from having charged a higher interest rate. Adjusting down payments offers far more protection than interest rate changes for risks that have a high potential of materializing in the first few years of a mortgage. Second, there are

[18] For example, in Bester, 1985's model of adverse selection, leverage is used as a screening device that leads to a separating equilibrium, with high risk types choosing higher leverage contracts, and then having to pay higher interest rates to compensate the lender for risk.

[19] In this vein, the papers that study this more complex setting tend to focus on down payment adjustments, finding substantial changes in quantities (Einav, Jenkins, and Levin, 2012; Gupta and Hansman, 2022). Evidence from other contexts supports the importance of adjustments in quantities and skin-in-the-game requirements for managing these collateral risks (e.g. Benmelech and Bergman, 2009; Capponi et al., 2020).

Fair Lending Laws that scrutinize adjustments through pricing (interest rates, points, and fees), which may make it easier to adjust mortgage contracts through down payments (FDIC, 2021).

Credit Supply and Demand: I now interpret the findings in light of credit supply and demand explanations. The first takeaway is that the LTV reduction in flood zones cannot be entirely explained by changing credit demand due to unobserved amenities, since the two identification strategies directly address this omitted variable. Amenities are unlikely to discontinuously change at the flood insurance cutoff, and amenities are held fixed as new maps are introduced. Under the identifying assumptions, these estimates thus represent the causal effect of flood risk on mortgage terms. However, there are two ways to interpret this causal effect. The first is that lenders screen on uninsurable flood risk (supply). The second is that uninsurable flood risk induces advantageous selection, whereby only those who can self-insure are willing to take on uninsurable flood risk (demand). A number of robustness tests speak against this demand-side interpretation of the estimates. First, the main specifications control for borrower incomes and credit scores, which are strongly correlated with capacity to self-insure. Second, loan-to-value ratios decline after the remapping specifically for under-insured borrowers that have higher credit risk (lower credit score and lower property values). Such borrowers are unlikely to have a higher capacity to self-insure, and in fact are likely to rely more on insurance payments to smooth shocks. Third, mortgage denials also increase after the remapping, suggesting a decline in credit supply since denials are at the discretion of the bank and not determined by the borrower (Gete and Reher, 2018). Taken together, these results support to a supply-side interpretation of the findings.

5. Conclusion

Climate change is likely to intensify flood damage in the years to come. Policymakers are concerned about what this means for the financial system. Despite the large potential losses, there is limited empirical evidence on who bears flood risk in residential mortgage markets.

This paper uses two sources of plausibly exogenous variation to show that lenders ration credit for mortgage borrowers that are likely to be under-insured against flood risk, requiring higher down payments. Tighter credit changes the composition of flood zones to higher income and higher credit quality individuals that are less likely to be liquidity constrained. These results suggest that limited flood insurance and credit rationing help to deter individuals with less risk-bearing capacity away from high risk areas.

There are several policy implications of these findings. The results suggest that fully insurable borrowers benefit most from the flood insurance program because they do not get credit rationed. The results suggest that in the absence of a government insurance program, all mortgage borrowers in flood zones would experience credit rationing. Furthermore, government flood maps are an important source of information for lenders. This shows the importance of the flood insurance and informational provision in supporting mortgage lending in high risk areas like flood zones.

Code Availability: The replication code is available in the Harvard Dataverse at https://dataverse.harvard.edu/dataset.xhtml?persistentId=doi:10.7910/DVN/WKAUFM.

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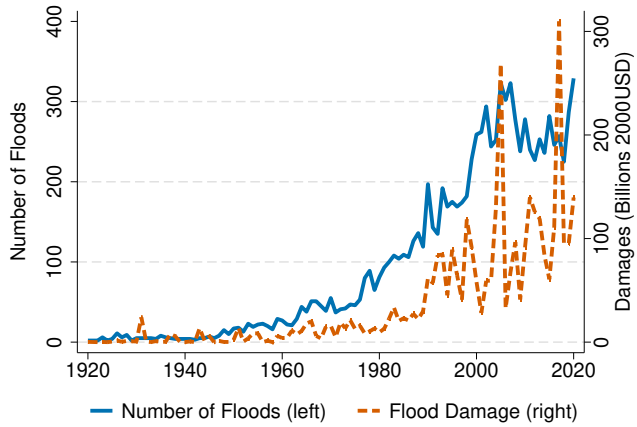
6. Figures and Tables

Figures

Figure 1
~~Worldwide Flood Disasters Over the Last Century~~

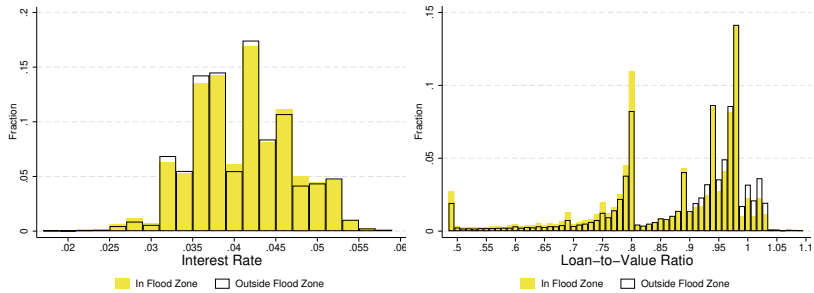


Source: EM–DAT International Disaster Database

~~*Notes:*~~ This graph plots a time series of the number of floods and total property damage caused by floods worldwide, as recorded in the Georeferenced Emergency Events Database (EM-DAT) atlas. To be recorded as a disaster in the database, an event must meet at least one of the following criteria: ~~ten~~ or more people reported killed, 100 or more people reported affected, a declaration of a state of emergency, and/or a call for international assistance. *Alt Text: Time-series graph showing the rise in the number of floods and flood damage over the last century worldwide.*

Figure 2
~~Histograms of Mortgage Characteristics by Flood Zone~~

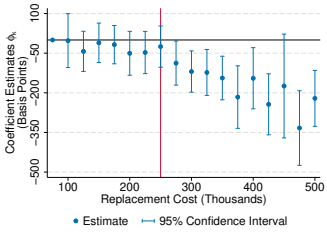
(a) Interest ~~R~~ates (b) Loan-to-~~Value Ratios~~



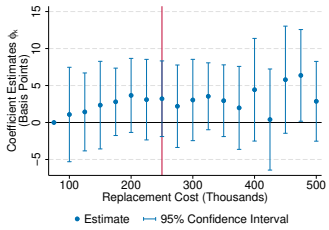
~~*Notes:*~~ This figure plots histograms of interest rates (~~Panel~~ A) and loan-to-value ratios (~~Panel~~ B) by FEMA flood zone status. *Alt Text: Histogram showing the distribution of interest rates and loan-to-value ratios inside and outside flood zones.*

Figure 3
~~Effect of Flood Zone on Mortgage Terms by Replacement Cost~~

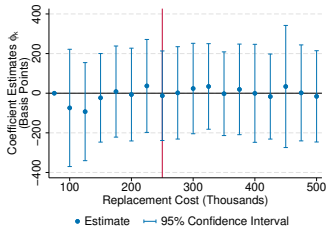
(a) LTV ~~Ratios (Basis Points)~~ Points~~)~~



(b) Interest ~~Rates (Basis~~

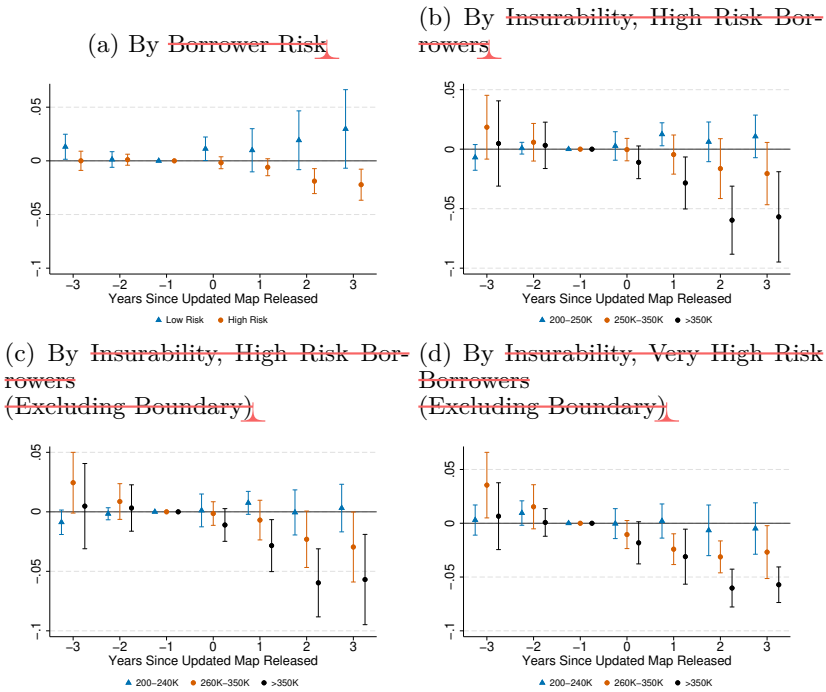


(c) Delinquency ~~(Basis Points)~~



~~*Notes:*~~ This figure plots regression coefficients of how the effect of FEMA flood zone classification on mortgages varies by the property's replacement cost. There are three outcome variables: LTV ratios (~~Panel~~ A), interest rates (~~Panel~~ B), and delinquency (~~Panel~~ C). *Delinquency* is a dummy variable that indicates whether a mortgage becomes delinquent for at least 30 days within the first three years of origination. Replacement costs are calculated as the product of the property's building size in square feet and construction costs for Florida, measured as dollars per square foot. Replacement costs are grouped into categories by increments of $25,000. Each dependent variable is regressed on a dummy variable indicating that the loan is in a flood zone interacted with a dummy for each replacement cost category. The category for replacement costs lower than $75,000 is omitted. All estimates can be interpreted as the effect of flood zone for that replacement cost category relative to the effect of the omitted category. The regression includes zip code-year fixed effects and a rich set of control variables which include the combined loan-to-value ratio for other liens on the property, property value, maturity, debt-to-income ratio, and dummy variables which indicate first mortgages, second homes, ~~low grade~~ mortgages, full document mortgages, jumbo loans, FHA/VA backed, and adjustable rate loans. The specification also controls for flood insurance take-up rates at the flood zone - ~~zip~~ code - year level. The ~~95 percent~~ confidence intervals are based on standard errors which are clustered at the county level. *Alt Text: Graph with three subfigures a-c showing the effect of flood zone on loan-to-value ratios, interest rates, and delinquencies as a function on replacement cost.*

Figure 4
Heterogeneous Effects of Updated Flood Maps on LTVs (percentage points)



~~*Notes:*~~ This figure shows heterogeneity in the effect of updated FEMA flood maps that expand flood zone boundaries on LTVs at origination. In ~~Panel~~ A, the sample is split into two groups based on borrower risk. ~~Low risk~~ borrowers are those with both high credit scores (above 700) and high house prices (above ~~$250K~~). ~~High risk~~ borrowers are those with either low house prices, low credit scores, or both. Panel B focuses on the high risk borrowers from ~~Panel~~ A, but then splits the sample into three groups based on replacement costs as indicated in the legend. Panel C shows the same splits as ~~Panel~~ B except the sample excludes properties that are within ~~$10K~~ of the flood insurance cap. Panel D limits the sample to very high risk borrowers who have both low house prices and low credit scores, and also excludes properties that are within ~~$10K~~ of the flood insurance cap. The figures report the coefficients from estimating a difference-in-differences regression that allows the effect to vary by year relative to the release of the updated flood map. All estimates can be interpreted as the effect relative to the year prior to the updated map. The ~~95 percent~~ confidence intervals are based on standard errors which are clustered at the county level. *Alt Text: Graph with four subfigures a-d showing the effect of flood zone remappings on loan-to-value ratios for different subsamples of high risk borrowers.*

Tables

Table 1
Summary Statistics

	FloodZone		NonFloodZone	
	mean	sd	mean	sd
Panel A: Loan Characteristics				
Credit Score	727.5	54.3	721.3	55.8
Applicant Annual Income ($000s)	100.0	143.4	76.3	84.9
Property Value ($000s)	278.7	291.4	205.7	150.7
LTV (%)	86.6	14.1	89.3	13.0
Interest Rate (%)	4.20	0.60	4.19	0.59
Delinquent Share (%)	3.07	17.3	3.79	19.1
Maturity (months)	354.4	31.0	354.8	30.0
DTI (%)	34.8	14.5	35.3	14.1
Combined LTV (%)	88.0	11.9	90.1	11.4
Observations	57,648		242,259	
Panel B: Flood Insurance Characteristics				
Takeup Rate (%)	26.2	18.5	6.80	8.84
Claim Probability (%)	1.51	4.29	1.13	2.91
Claim ($000s)	13.6	15.3	14.0	17.5
Assessed Building Value ($000s)	193.7	220.5	195.6	198.9
Observations (~~Zip-Year~~)	5,397		5,397	

~~*Notes:*~~ This table provides summary statistics on mortgage and flood insurance characteristics for the estimation sample and provides a breakdown by FEMA flood zone status. Panel A focuses on mortgage characteristics. Panel B focuses on flood insurance. The policy-level data from FEMA on flood insurance is anonymized, but includes identifying information about ~~zip~~ code, flood zone classification, and year in force. Take up rates are calculated each for ~~zip~~ code - year combination, inside of flood zones and outside of flood zones. This uses Zillow ZTRAX property assessment data to obtain counts of the number of housing units in and out of flood zones. This table presents the overall average and standard deviation for those ~~zip~~ code - ~~year level~~ observations. ~~*Alt Text: Table with summary statistics on loan characterstics and flood insurance characteristics for two samples, borrowers inside a flood zone and outside a flood zone.*~~

Table 2
~~**Descriptive Facts on How Mortgage Terms Vary by Flood Zone**~~

	(1) LTV Ratio	(2) Interest Rate	(3) Delinquency
~~*Panel A: Excludes Controls and Zip-Year Fixed Effects*~~			
FloodZone	-0.0268***	0.0000	-0.0072**
	(0.0054)	(0.0003)	(0.0035)
Adjusted R-Squared	0.01	0.00	0.00
~~*Panel B: Excludes Controls, Includes Zip-Year Fixed Effects*~~			
FloodZone	-0.0187***	-0.0002***	-0.0040***
	(0.0023)	(0.0000)	(0.0010)
Adjusted R-Squared	0.13	0.42	0.06
~~*Panel C: Includes Controls, Excludes Zip-Year Fixed Effects*~~			
FloodZone	-0.0087*	0.0007***	0.0014
	(0.0047)	(0.0002)	(0.0023)
Adjusted R-Squared	0.19	0.09	0.05
~~*Panel D: Includes Controls and Zip-Year Fixed Effects*~~			
FloodZone	-0.0083***	0.0001***	-0.0004
	(0.0016)	(0.0000)	(0.0008)
Adjusted R-Squared	0.25	0.45	0.09
Observations	299,907	299,907	299,907

~~*Notes:*~~ This table reports regression coefficients of the effect of FEMA flood zone status on mortgage characteristics. The three dependent variables are the mortgage's loan-to-value (LTV) ratio, interest rate, and delinquency. *Delinquency* is a dummy variable that indicates whether the mortgage becomes more than ~~30-days~~ delinquent within the first three years of origination. *FloodZone* is a dummy variable which indicates whether the mortgage is located in a FEMA flood zone when it was originated. Control variables include the borrower's credit score, income, and property value. Fixed effects and control variables are only included where indicated. Standard errors are reported in parentheses and are clustered at the county level. Significance Levels: * (p<0.10), ** (p<0.05), *** (p<0.01). ~~*Alt Text: Regression table with coefficient estimates for the effect of flood zone on mortgage terms. The table has three columns for each mortgage outcome, loan-to-value ratio, interest rate, and delinquency.*~~

Table 3
~~**Effect of Capped Flood Insurance on Mortgages in Flood Zones**~~

	(1)	(2)	(3)	(4)
~~*Panel A: LTV Ratio*~~				
CapBinds	-0.0465***	-0.0366***	0.0072***	0.0053***
	(0.0035)	(0.0024)	(0.0013)	(0.0012)
FloodZone	-0.0190***	-0.0117***	0.0002	-0.0010
	(0.0052)	(0.0024)	(0.0019)	(0.0020)
CapBinds × FloodZone	-0.0139***	-0.0113***	-0.0085***	-0.0081***
	(0.0043)	(0.0030)	(0.0019)	(0.0019)
Adjusted R-Squared	0.0383	0.1448	0.4544	0.4756
Observations	299,907	299,907	299,907	299,907
~~*Panel B: Interest Rate*~~				
CapBinds	-0.0018***	-0.0012***	-0.0001**	-0.0001**
	(0.0001)	(0.0001)	(0.0000)	(0.0000)
FloodZone	0.0002	-0.0001	0.0000	0.0000
	(0.0003)	(0.0000)	(0.0000)	(0.0001)
CapBinds × FloodZone	-0.0003**	-0.0002***	0.0000	0.0000
	(0.0001)	(0.0001)	(0.0000)	(0.0000)
Adjusted R-Squared	0.0227	0.4257	0.5600	0.5785
Observations	299,907	299,907	299,907	299,907
~~*Panel C: Delinquency*~~				
CapBinds	-0.0244***	-0.0134***	-0.0016**	-0.0008
	(0.0025)	(0.0017)	(0.0007)	(0.0006)
FloodZone	-0.0077*	-0.0046***	-0.0011	-0.0007
	(0.0043)	(0.0012)	(0.0015)	(0.0016)
CapBinds × FloodZone	0.0046	0.0037***	0.0015	0.0004
	(0.0033)	(0.0013)	(0.0012)	(0.0011)
Adjusted R-Squared	0.0038	0.0640	0.1416	0.1573
Observations	299,907	299,907	299,907	299,907
~~Zip-Year FE~~	N	Y	Y	Y
Controls	N	N	Y-restricted	Y-full

~~*Notes:*~~ This tables reports estimates of the effect of FEMA flood zone classification on mortgages varies by the whether the flood insurance cap binds. There are three outcome variables: LTV ratio (~~Panel~~ A), interest rate (~~Panel~~ B), and delinquency (~~Panel~~ C). Delinquency is defined as whether the mortgage becomes more than ~~30-days~~ delinquent within the first ~~three~~ years of origination. *FloodZone* is a dummy that indicates if a mortgage is located in a FEMA flood zone. *CapBinds* is a dummy that indicates whether the replacement cost exceeds the flood insurance cap of $250,000. ~~Zip~~ code-year fixed effects are included where indicated. In column ~~(4)~~, the regression includes the full set of controls: flood insurance take-up rates (~~zip-~~flood zone -year), the borrower's credit score, annual income, combined LTV for other liens on the property, property value, maturity, DTI ratio, and dummy variables which indicate first mortgages, second homes, ~~low grade~~ mortgages, full document mortgages, jumbo loans, and adjustable rate loans. Column ~~(3)~~ includes all the same controls as ~~(4) except~~ income and credit score. Standard errors are reported in parentheses and are clustered at the county level. Significance Levels: *

~~(p<0.10), ** (p<0.05), *** (p<0.01). *Alt Text: Regression table with coefficient estimates for the effect of uninsurable flood risk in flood zones on three mortgage variables: for loan-to-value ratio, for interest rate, for delinquency.*~~

Table 4
~~Descriptive Statistics by Remapping Year for Loans Originated in 2010~~

	Year of ~~R~~emapping:						
	2012		2013		2014		P-
	Mean	SD	Mean	SD	Mean	SD	Value
~~*Panel A: Loan Characteristics*~~							
Number of Loans	2,314		3,015		2,575		.
LTV (%)	89.1	15.4	91.7	11.7	88.1	14.8	.12
Interest Rate (%)	4.94	.457	4.91	.474	4.88	.48	.22
Loan Amount ($000s)	147	102	145	84.2	150	96.4	.96
Property Value ($000s)	176	152	163	116	179	137	.84
Maturity (months)	356	27.1	356	26.9	355	29.5	.56
Credit Score	714	58.9	702	59.9	718	61.6	.25
Annual Income ($000s)	67.3	81.6	64.7	66.8	67.6	60.2	.96
Combined LTV (%)	91	12.1	94.1	10.3	91.1	12.3	.02
Replacement Cost ($000s)	22.6	7.65	21.7	7.99	21.7	8.05	.78
Building Size (Square Feet)	1913	648	1837	677	1839	682	.78
Jumbo Loan Share (%)	1.17	10.7	.564	7.49	1.05	10.2	.64
~~*Panel B: Flood Insurance Characteristics*~~							
Number of ~~Zip Codes~~	59		54		52		.
SFHA Takeup (%)	19.2	11.8	25.6	12.1	33.9	16.4	.11
Non-SFHA Takeup (%)	7.58	6.69	4.68	3.41	7.38	5.98	.14
Average Claim ($000s)	17.6	11.3	21.6	15.6	21.8	13.9	.36
~~*Panel C: Other Zip Characteristics*~~							
ACS Housing Units (000s)	11.1	5.01	13.6	6.22	12.1	5.58	.19
ACS Population Growth (%)	.056	.119	.104	.298	.0452	.155	.28
HMDA Denial Rate (%)	22.4	4.65	22.9	6.61	20.6	5.16	.21

~~*Notes:*~~ This table presents average loan-level and ~~zip~~-level characteristics for mortgages originated in 2010. The sample is restricted to those ~~zip~~ codes which subsequently receive an updated map that expands flood zone boundaries. Mortgages are grouped by which year the associated county receives an updated map, and statistics on the average and standard deviation are presented. The last column indicates the ~~p-value~~ for the joint test that the means for each remapping year are the same. The acronym "SFHA" refers to special flood hazard areas, which are the official names for the FEMA-defined flood zones. ~~*Alt Text: Descriptive statistics for loan characteristics, flood insurance characteristics, and zip-level characteristics for zip codes by the year they receive their updated flood map.*~~

Table 5
~~**Dynamic Effects of Updated Flood Maps on Mortgages**~~

	LTV				Log Income	Credit Score
	(1)	(2)	(3)	(4)	(5)	(6)
Years ~~Since Updated Map~~:						
[1em] -3	-0.000995	-0.00146	0.00124	-0.000202	-0.0102	-0.641
	(0.00480)	(0.00364)	(0.00409)	(0.00342)	(0.0493)	(1.583)
-2	0.00111	0.000580	0.00218	0.00120	-0.0106	-0.443
	(0.00203)	(0.00216)	(0.00194)	(0.00213)	(0.0202)	(1.040)
-1	0	0	0	0	0	0
	(.)	(.)	(.)	(.)	(.)	(.)
0	0.000351	0.00123	-0.000156	0.000923	0.0112	0.760
	(0.00270)	(0.00299)	(0.00260)	(0.00282)	(0.00664)	(0.988)
1	-0.00483	-0.00244	-0.00446	-0.00233	0.0519***	3.271*
	(0.00316)	(0.00323)	(0.00337)	(0.00335)	(0.0160)	(1.628)
2	-0.0155***	-0.0106**	-0.0135***	-0.00968**	0.0935***	6.846***
	(0.00506)	(0.00440)	(0.00470)	(0.00428)	(0.0329)	(1.684)
3	-0.0202***	-0.0141**	-0.0172***	-0.0126**	0.147**	8.769***
	(0.00602)	(0.00558)	(0.00565)	(0.00545)	(0.0541)	(2.690)
Adjusted R-Squared	0.0689	0.1819	0.0766	0.1842	0.1179	0.0526
Observations	55,819	55,819	55,819	55,819	49,559	47,863
County FE	Y	Y	Y	Y	Y	Y
Year FE	Y	Y	Y	Y	Y	Y
Borrower Controls	N	Y	N	Y	N	N
Flood Insurance Controls	N	N	Y	Y	N	N

~~*Notes:*~~ This table reports estimates of the effect of updated FEMA flood maps that expand flood zone boundaries on LTV ratios (~~Columns~~ 1-4), log income (~~Column~~ 5), and credit score (~~Column~~ 6). The table reports the coefficients from estimating a difference-in-differences regression that allows the effect to vary by year relative to the release of the updated flood map. Estimates were constructed by regressing each dependent variable on a series of event-time dummy variables indicating the year relative to the release of the updated map. Relative year zero is the year that the map was released. The regression also includes year fixed effects and county fixed effects. All estimates can be interpreted as the effect relative to the year prior to the updated map. Columns ~~(2), (3), and (4)~~ include control variables as indicated. Borrower controls include loan-level controls for income and credit score. Flood insurance is calculated as the take-up rates of flood insurance within a flood zone-~~zip~~-year cell. ~~bservations~~ may differ because not all loans contain data on all variables. Standard errors are clustered at the county level. Significance Levels: * (p<0.10), ** (p<0.05), *** (p<0.01). ~~*Alt Text: Regression estimates of dynamic treatment effects for the release of updated flood maps on mortgage loan-to-value ratios, borrower incomes, and borrower credit scores.*~~

Table 6
~~**Pooled Effect of Updated Maps on Mortgages**~~

	Interest Rate (1)	Delinquency (2)	DTI (3)	Maturity (4)	Dollar per Sq.Ft. (5)
Post	0.0006	-0.0098***	-0.2720	-0.0548	7.1418
	(0.0007)	(0.0024)	(0.2529)	(0.3786)	(7.2821)
Adjusted R-Squared	0.39	0.05	0.05	0.00	0.01
County FE	Y	Y	Y	Y	Y
Year FE	Y	Y	Y	Y	Y
Observations	55,692	55,819	19,505	55,819	51,820

~~*Notes:*~~ This table reports estimates of the effect of updated flood maps that expand flood zone boundaries on interest rates, delinquencies, debt-to-income ratios, maturity, and property value per square foot. It reports the coefficients from a difference-in-differences regression, where the effect varies by the year relative to the release of the updated flood map. $InterestRate$ is the mortgage's interest rate at origination, measured in percentage points. $Delinquency$ is a dummy variable that indicates whether a mortgage becomes more than ~~30-days~~ delinquent within the first three years of origination. DTI is the mortgage's debt-to-income ratio at origination. $Maturity$ is the mortgage's maturity at origination, measured in months. *Dollar per* ~~*Sq.Ft.*~~ is the property value at origination divided by the total size of the property in square feet. *Post* is a dummy variable that indicates whether that mortgage is originated on or after the introduction of the updated flood map. Post equals zero in the pre-remapping period. Observations may differ because not all loans contain data on all variables. The regression also includes year fixed effects and county fixed effects. Standard errors are reported in parentheses and are clustered at the county level. Significance Levels: * (p<0.10), ** (p<0.05), *** (p<0.01). ~~*Alt Text: Regression estimates of treatment effects for the release of updated flood maps on mortgage interest rates, delinquency, debt-to-income ratios, maturity, and property value per square foot.*~~

Table 7
~~Pooled Effect of Updated Maps by whether the NFIP Cap Binds~~

	LTV Ratio		Interest Rate	
	(1)	(2)	(3)	(4)
Post=1	0.0105***	0.0103***	0.0005	0.0005
	(0.0028)	(0.0034)	(0.0006)	(0.0006)
Post=1 × CapBinds=1	-0.0081*	-0.0066*	-0.0003**	-0.0002*
	(0.0046)	(0.0038)	(0.0001)	(0.0001)
Adjusted R-Squared	0.10	0.20	0.47	0.49
County FE	Y	Y	Y	Y
Year FE	Y	Y	Y	Y
Replacement Cost Control	Y	Y	Y	Y
Other Controls	N	Y	N	Y
Observations	51,595	51,595	51,595	51,595

~~*Notes:*~~ This table reports estimates of the effect of updated flood maps that expand flood zone boundaries on mortgage terms by whether or not the flood insurance coverage limit binds. The two dependent variables are loan-to-value ratio at origination (LTV) and interest rate at origination. *Post* is a dummy variable that indicates whether that mortgage is originated on or after the introduction of the updated flood map, and zero otherwise. *CapBinds* is a dummy variable which equals 1 if the replacement cost of the home is above the $250,000 NFIP flood insurance coverage limit, and 0 otherwise. The regression also includes year fixed effects and county fixed effects. All specifications include loan-level controls for replacement cost. Columns ~~(2) and (4)~~ also include the mortgage's debt-to-income ratio and credit scores as loan-level controls. Standard errors are clustered at the county level. Significance Levels: * (p<0.10), ** (p<0.05), *** (p<0.01). ~~*Alt Text: Regression estimates of treatment effects for the release of updated flood maps by whether the flood insurance cap binds on mortgage loan-to-value ratios and interest rates.*~~